

unbound: Python module loaded unconditionally regardless of DNSBL configuration state

Component: Services: Unbound DNS

Version: OPNsense 26.1.5 (confirmed present in master as of 2026-03-27)

Severity: High — complete DNS resolution failure on systems without active DNSBL blocklists

File: src/etc/inc/plugins.inc.d/unbound.inc

Line: 206

Summary

In `unbound_generate_config()`, line 206 unconditionally initializes `$module_config` with the string 'python ', causing Unbound to always load the Python interpreter module regardless of whether any DNSBL blocklists are enabled. On systems where no blocklists are configured, the Python module loads unnecessarily, and in chroot environments where Python's runtime dependencies are unavailable, this causes Unbound to fail silently or return `SERVFAIL/NXDOMAIN` for all queries.

Environment

OPNsense 26.1.5 (FreeBSD 14) | Unbound 1.24.2 | DNS Forwarding mode | No DNSBL blocklists configured | Five-zone VLAN architecture

Steps to Reproduce

1. Install OPNsense 26.1.5 fresh
2. Enable Unbound DNS with Query Forwarding
3. Do NOT configure any DNSBL blocklists
4. Restart Unbound: `configctl unbound restart`
5. Inspect `/var/unbound/unbound.conf` for module-config line

Expected Result

```
module-config: "iterator"
```

Actual Result

```
module-config: "python iterator"
python-script: unbound-dnsbl/dnsbl_module.py
```

Root Cause

```
Line 206 in unbound_generate_config():
```

```
    $module_config = 'python ';
```

This initialization is unconditional. There is no check for whether any DNSBL blocklists are enabled before this line.

Impact

On systems without DNSBL configured: Python module loads unnecessarily, consuming memory. In chroot environments where Python dependencies are unavailable: Unbound returns SERVFAIL or NXDOMAIN for all forwarded queries. The `dnsbl_module.py` script is always deployed and referenced even when it serves no purpose.

Fix

```
$dnsbl = config_read_array('OPNsense', 'unboundplus', 'dnsbl');
$has_active_dnsbl = false;
if (!empty($dnsbl['blocklist'])) {
    foreach ($dnsbl['blocklist'] as $bl) {
        if (!empty($bl['enabled'])) { $has_active_dnsbl = true; break; }
    }
}
$module_config = $has_active_dnsbl ? 'python ' : '';
```

Validation

After fix, with no DNSBL blocklists configured:

```
$ grep 'module' /var/unbound/unbound.conf
module-config: "iterator"
```

Unbound starts cleanly with iterator-only module chain.

Suggested Commit Message

unbound: load python module only when DNSBL blocklists are active